

01

Foundations & Systematic Assessment

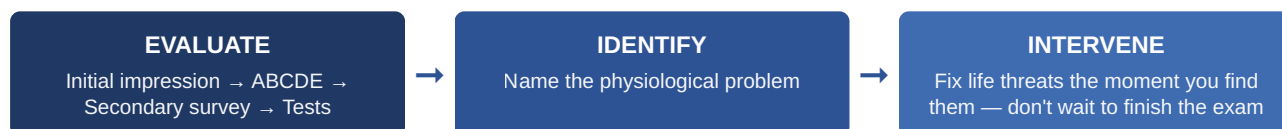
Master systematic assessment (PAT → ABCDE → Secondary Survey → Early Escalation) and recognize the sick child within the first 5 minutes.

1. Foundations & Systematic Assessment

Learning Objectives

- Apply the Evaluate → Identify → Intervene sequence to every child on first contact.
- Form an accurate Initial Impression using the Paediatric Assessment Triangle (PAT) before touching the child.
- Complete a structured ABCDE primary assessment and recognise the five-point breathing and circulation checks.
- Take a focused SAMPLE history and apply the DIRECT approach to organise a bedside handover.
- Assign an accurate triage acuity level and know when to escalate immediately.

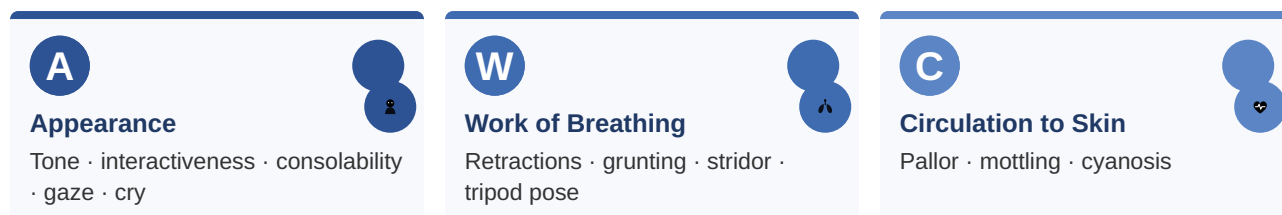
1. The Loop: Evaluate → Identify → Intervene



WHY THIS MATTERS

Out-of-hospital arrest survival ≈10%; in-hospital ≈30%. Respiratory & circulatory failure is usually preventable if caught in the first 5 minutes.

2. Initial Impression — PAT (from the doorway, hands off)



2 abnormal limbs → sick child, urgent primary assessment. **3 abnormal** → impending cardiopulmonary failure.

3. Primary Assessment — ABCDE

A
Airway

Talking/crying normally? No gurgle/stridor? Check head position, suction needs.

B
Breathing

5-point check → rate, air entry, sounds, effort, SpO₂ (see box below)

C
Circulation

5-point check → pulse, volume, CRT, temp/colour, BP

D
Disability

AVPU · pupils · glucose in every drowsy child

E
Exposure

Fully undress — look for bruising/petechiae; prevent hypothermia


B — Breathing, 5-Point Check

- 1 Rate for age — >60 or <10/min = danger
- 2 Equal air entry
- 3 Sounds: stridor / wheeze / grunt
- 4 Effort: flaring, recession, see-saw
- 5 SpO₂ <94% abnormal


C — Circulation, 5-Point Check

- 1 Pulse rate/rhythm — <60/min ominous
- 2 Pulse volume — central-peripheral gap = earliest shock sign
- 3 CRT — press 5s, >2s prolonged
- 4 Temp/colour/mottling
- 5 BP — remember **60–70–80–90** (min systolic by age)


D & E IN ONE LINE

AVPU: Alert → Voice → Pain → Unresponsive — P/U mandates emergency referral. Check glucose (<60 mg/dL = treat now) and fully expose while preventing hypothermia.

4. Secondary Assessment — SAMPLE

S
Signs & Symptoms

Of the presentation

A
Allergies
M
Medications

Given so far

P
Past History

Medical background

L
Last Meal

Timing

E
Events

Leading up to presentation

Follow with a focused head-to-toe exam. Basic work-up: glucose, CBC, CRP, electrolytes, LFT/RFT, gas + lactate; CXR if respiratory; ECG/ECHO if cardiac; USG if abdo/renal concern.

5. Triage — Assigning Acuity

Level	Category	Example	Immediate action
1	Resuscitation	Arrest, active seizure, hypotensive shock	CPR, intubation
2	Emergent	Compensated shock, distress	O ₂ , NS bolus
3	Urgent	Dehydration, post-ictal, pain	Monitor, treat

4	Less urgent	Stable, minor injury	Can wait
5	Non-urgent	Minor illness	Routine

1 Visual assessment
First glance from across the room



2 Chief complaint + SAMPLE
Focused history



3 Focused ABCD
Quick targeted exam



4 Form a hypothesis
What's most likely going on?



5 Assign acuity
Level 1–5



6 Reassess
Re-triage if condition changes while waiting

6. DIRECT — Bedside Handover



D-I-R

Presentation given · Interventions already done ·
Response to treatment so far



E-C-T

Evolving concerns · Care plan · Tasks pending for next
shift

Use this structure at every shift change and escalation call so nothing is lost in translation.

7. Ready the Environment



START-OF-SHIFT CHECKLIST

O₂ supply · suction · infusion lines · ventilator/BiPAP/HFNC · monitor alarms · crash cart layout
by heart (airway / breathing / circulation-access / drugs / defib-ECG / misc) · Broselow tape
ready for unknown-weight children.

**NURSE SELF-CARE**

A calm, rested, focused nurse is more effective than one rushing in drained. Your own wellbeing is foundational to the child's safety.

Compiled & adapted from

- PEERS Manual (2025) — Section 1: Recognition of a Sick Child; Section 2: Principles of Triage (pp. 4–14, 12–14, Appendix vitals tables).
- BPCCN Guide — Daily Routine of PICU Nurses; Identification of a Sick Child; Triage and its Levels in Paediatric Emergency (pp. 2–22).

Page-level citations to be added in a subsequent pass. Content has been edited and condensed for print/online study use; refer to the full source manuals for exhaustive detail.

02

Respiratory Care

Differentiate respiratory distress from respiratory failure and initiate safe airway support, including non-invasive respiratory support before intubation.

2. Respiratory Care

Learning Objectives

- Classify a respiratory presentation as upper airway obstruction, lower airway obstruction, lung tissue disease, or disordered control of breathing.
- Select the correct oxygen delivery device using the AIM framework (Assess, Identify, Match).
- Perform safe two-person bag-mask ventilation and initiate HFNC.
- Deliver a high-quality nebulisation and titrate bronchodilator therapy for acute wheeze.
- Recognise when non-invasive support is failing and escalation to invasive ventilation is required.

1. Distress vs Failure — Classify the Pattern



Upper Airway Obstruction

Foreign body, croup, epiglottitis. Signs: stridor, voice change, drooling/gurgling, poor chest rise.



Lower Airway Obstruction

Asthma, bronchiolitis. Signs: ↑RR & effort, wheeze, cough, ↓air movement.



Lung Tissue Disease

Pneumonia, oedema, ARDS. Signs: tachypnoea, grunting, crackles, hypoxaemia.



Disordered Control

CNS infection, TBI, sedative overdose. Signs: irregular rate, shallow breaths, apnoea.



ESCALATE IMMEDIATELY

Airway procedures generate aerosol — use PPE. A child with risk factors needs close monitoring or early referral before distress becomes failure.

2. First Steps — Open & Support the Airway

1

Open the airway

Correct alignment; head-tilt–chin-lift, or jaw-thrust if spine injury suspected



2

Adjunct if needed

Oropharyngeal if unresponsive/no gag; nasopharyngeal if semi-conscious (avoid in facial/skull injury)



3

Tracheostomy in situ?

Suction, or change tube if blockage not clearing with suction



4

Check SpO₂

<94% on air → supplemental O₂, titrate to 95–98% (100% only in shock via NRM)

3. Oxygen Devices — the AIM Framework

A
Assess

Patient need

I
Identify

Technology available

M
Match

Technology to need

Device	Flow	FiO ₂	Notes
Nasal prongs	≤4 L/min	Low, variable	Mild hypoxia
Simple face mask	6–10 L/min	Variable	Flow < minute ventilation
Non-rebreathing mask	6–10 L/min	>95%	Sick child / shock
HFNC	Up to 2 L/kg/min	0.21–1.0	Some PEEP effect, dead-space washout

Breathing still inadequate despite O₂? Support with positive-pressure bag-mask breaths — a core peri-arrest skill.

4. Bag-Mask Ventilation (BMV)


Good Seal

Age-appropriate mask; allow full inspiratory time to avoid breath-stacking


2-Person Technique

One holds E-C clamp seal, one squeezes the bag — use when seal is hard alone


Escalate

LMA if competent; intubation only per RSI principles by a trained provider

5. Non-Invasive Ventilation & HHHFNC

Type	Definition	Common causes
I — Hypoxaemic	PaO ₂ <60, normal/low CO ₂	Pneumonia, mild ARDS, oedema, bronchiolitis
II — Hypercapnic	PaCO ₂ >50, hypoxaemia common	Impaired drive, neuromuscular weakness, chest wall issue


WHEN NIV IS FAILING

Worsening work of breathing, rising CO₂, or falling consciousness → escalate promptly to invasive ventilation (Section 12). Also useful in shock/cardiac failure to offload the work of breathing.

6. Pneumonia — Quick Bundle



Monitor

Continuous cardio-resp + SpO₂



Support

Position, clear airway, high-flow O₂,
bronchodilator if needed



Treat

Antipyretics; antibiotics within 1st
hour; consider fluids

7. Acute Wheeze — Drug Doses



NEBULISED SALBUTAMOL

0.15 mg/kg (min 2.5, max 5 mg) every 20 min ×3, then continuous 0.5 mg/kg/hr if needed. ±
Budesonide/Ipratropium (250 mcg <20 kg; 500 mcg >20 kg). O₂-driven at 6–8 L/min —
recheck SpO₂, RR, distress every few minutes.

Adjunct	Dose
Prednisolone	2 mg/kg PO, then 1 mg/kg OD
Dexamethasone	0.3–0.6 mg/kg (max 12–16 mg/day)
Methylprednisolone	1–2 mg/kg/day IV
Magnesium sulfate	50–100 mg/kg over 0.5–1 hr



LINKS TO THE HANDS-ON LAB

Oxygen-device selection stations, HFNC setup, nebuliser technique, and two-person BMV
practice — see the Workshop Schedule.

Compiled & adapted from

- PEERS Manual (2025) — Section 3: Basic Management & O₂ Delivery Devices; Section 4: Management of Respiratory Emergencies (pp. 15–30).
- BMV Workshop Manual (PediCritCon 2025) — Non-Invasive Ventilation & HHHFNC (pp. 6–18).

Page-level citations to be added in a subsequent pass. Content has been edited and condensed for print/online study use; refer to the full source manuals for exhaustive detail.

03

Shock, Hemodynamics, Sepsis & Cardiac ICU Care

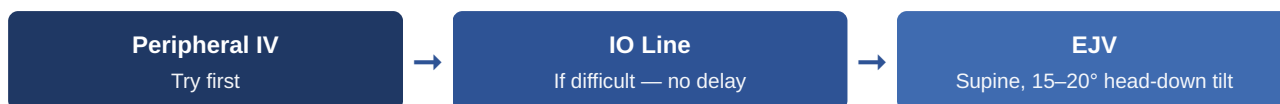
Classify shock and initiate fluid/vasoactive therapy safely; recognize sepsis and anaphylaxis early; apply the 4Ds/ROSE framework to fluid therapy; manage hemodynamic instability in the post-operative cardiac surgical child.

3. Shock, Hemodynamics, Sepsis & Cardiac ICU Care

Learning Objectives

- Classify shock by type and severity and secure timely vascular access (peripheral, IO, EJV).
- Run the 1-hour sepsis bundle and recognise anaphylaxis, giving IM adrenaline without delay.
- Apply the 4Ds/ROSE framework to fluid resuscitation, maintenance, and de-escalation.
- Recognise fluid overload and know when to restrict or actively remove fluid.
- Recognise low cardiac output syndrome and pulmonary hypertensive crisis in the post-cardiac-surgery child.

1. Vascular Access



Give adequate analgesia unless comatose. EJV avoided in: neck trauma/infection, thrombosis/vascular anomaly, unclear cervical spine.

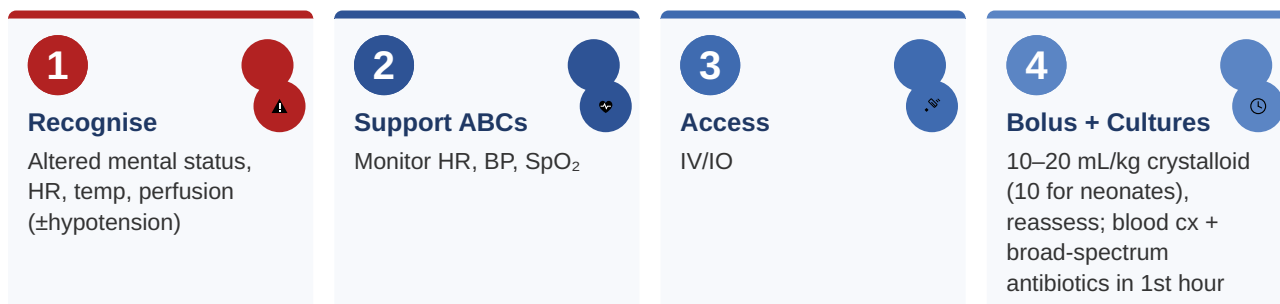
2. Classifying & Treating Shock



FLUID BOLUS

Normal saline 10–20 mL/kg IV — reassess after each bolus before repeating. Track BP (correct cuff size), urine output, ultrasound/lactate if available.

3. The 1-Hour Sepsis Bundle



4. Anaphylaxis



ADRENALINE 1:1000 (1 MG/ML, UNDILUTED)

IM 0.01 mg/kg every 15 min ×2, then every 4 hr PRN. Give without delay — do not wait for IV access.

5. Fluid Therapy — the 4Ds & ROSE

D

Drug

Fluid IS the drug

D

Dose

How much

D

Duration

How long

D

De-escalation

When to stop/remove

RESUSCITATION

Minutes · severe shock · fast, reassess · +ve balance expected



OPTIMISATION

Hours · unstable but not hypovolaemic · neutral balance



STABILISATION

Days · maintenance · normal lactate/vitals · feeds established



EVACUATION

Days–weeks · mobilise excess fluid · –ve balance targeted

Speed rule: Resuscitation boluses (10–20 mL/kg) — FAST, reassess. Rehydration/maintenance — SLOW. Sick children over-secrete ADH → restrict total intake to 50–70% of standard maintenance until feeds established.



FLUID OVERLOAD CASCADE

Excess IV fluid → interstitial leak → visceral oedema → intra-abdominal hypertension → venous hypertension → renal congestion → oliguria, low Na, SIADH. Ask every round: do I need to give fluid, or restrict/remove it?

Fluid Overload % = (Intake – Output) ÷ Baseline weight × 100. >10% is clinically significant. Track hydration, perfusion, and lactate daily.



TAKE-HOME

Isotonic fluids unless specified · crystalloids before colloids · fluid can be "poisonous" · restrict maintenance to 50–70% · Resuscitation FAST, Replacement SLOW, Maintenance SLOWER · monitor regularly.

6. Post-Op Cardiac ICU Hemodynamics

Instability peaks immediately post-op. On PICU arrival: swift, safe transfer of monitoring/infusions ("Formula 1 tyre change"), assess stability, structured handover — teamwork is everything.



WATCH FOR AT HANDOVER

Monitor/pump battery not swapped · infusion pump failure stopping critical drips · inadequate sedation causing agitation · monitor switched off mid-transfer.

First samples: ABG + ACT (ventilation, lactate, K⁺, Hb/bleeding). CXR, echo and ECG mandatory in every post-op cardiac child.

**Low Cardiac Output Syndrome**

Feeble/absent pulses, poor perfusion, rising lactate, falling urine output. Escalate before hypotension appears — it's a late sign.

**Pulmonary HTN Crisis**

After high-pulmonary-flow lesion repair: sudden desat, right heart strain, collapse. Avoid hypoxia/hypercarbia/acidosis triggers.

**Ductal-Dependent Lesions**

Keep Prostaglandin E1 (PGE1) ready and running to keep the ductus open until repair/palliation.

**LINKS TO THE HANDS-ON LAB**

Fluid-calculation workshop, timed sepsis-bundle drill, and cardiac hemodynamics case stations (low cardiac output, tamponade) — see the Workshop Schedule.

Compiled & adapted from

- PEERS Manual (2025) — Section 5: Management of Circulatory Problems (pp. 31–46).
- FLUID Therapy in PICU (PediCritiCon 2025 lecture) — 4Ds, ROSE principle, fluid overload monitoring.
- PeC3N — Nursing Care in the Immediate Postoperative Period, Pediatric Cardiac ICU.

Page-level citations to be added in a subsequent pass. Content has been edited and condensed for print/online study use; refer to the full source manuals for exhaustive detail.

04

Arrhythmias, BLS & Resuscitation (PALS)

Identify pediatric arrhythmias, including common post-cardiac-surgery rhythms, and deliver safe CPR and defibrillation.

4. Arrhythmias, BLS & Resuscitation (PALS)

Learning Objectives

- Distinguish shockable (VF, pulseless VT) from non-shockable (asystole, PEA) arrest rhythms and treat accordingly.
- Recognise sinus tachycardia, SVT, and ventricular tachycardia by rate, regularity and QRS width.
- Prepare and administer emergency arrhythmia drugs at correct pediatric doses.
- Deliver high-quality single- and multi-rescuer CPR and operate a defibrillator/AED safely.
- Recognise arrhythmias specific to the post-cardiac-surgery child (junctional ectopic tachycardia, complete heart block).

1. Arrest Rhythms — Shockable or Not?



Asystole

No pulse · NOT shockable · CPR + epinephrine



PEA

No pulse · NOT shockable · CPR + epinephrine + treat cause



VT / VF (pulseless)

No pulse · SHOCKABLE · CPR + defib + epi + amiodarone

- 1 Assess ABCs
- ↓
- 2 Monitor artefact vs true rhythm?
- ↓
- 3 Call for help / Code Blue
- ↓
- 4 Start CPR if no pulse
- ↓
- 5 Prepare defib / cardioversion
- ↓
- 6 Give drugs per PALS

2. Rhythm Recognition

Rhythm	QRS	Regular?	Pulse?	Initial Rx
--------	-----	----------	--------	------------

Sinus tachycardia	Narrow	Yes	Yes	Treat cause
SVT	Narrow	Yes	Yes	Vagal → adenosine → cardioversion
VT (with pulse)	Wide	Yes	Yes	Amiodarone / sync cardioversion
VT/VF (no pulse)	Wide	No	No	CPR + defib + epi
Bradycardia	Narrow	Regular	Y/N	Atropine ± CPR

**ANY NEW RHYTHM — BEDSIDE CHECKLIST**

1) Sinus? Every QRS after a normal P. 2) Rate, QRS width, PR, regularity. 3) Pulses/SpO₂/hemodynamics first, whatever the rhythm. 4) Correct acidosis, sugar, temp, electrolytes (Ca/K/Mg). 5) Know your drugs. 6) Defibrillator ready.

3. Emergency Drugs

Drug	Indication	Dose
Epinephrine	Arrest	0.01 mg/kg IV/IO q3–5min (1:10,000)
Adenosine	SVT	0.1 → 0.2 → 0.3 mg/kg rapid push (max 6/12 mg) + saline flush
Amiodarone	VT, VF	5 mg/kg IV over 20–60 min
Atropine	Bradycardia	0.02 mg/kg (min 0.1 mg) IV/IO

**DRAWING UP ADRENALINE 1:10,000**

Supplied 1 mg/mL (1:1000). Dilute 1 mL drug + 9 mL NS → 1 mL = 0.1 mg. Dose 0.01 mg/kg IV/IO = 0.1 mL/kg of diluted solution.

4. Basic Life Support

**Start**

Compressions
within 10 sec

**Push**

Hard & fast, 100–
120/min

**Avoid**

Hyperventilation

**Allow**

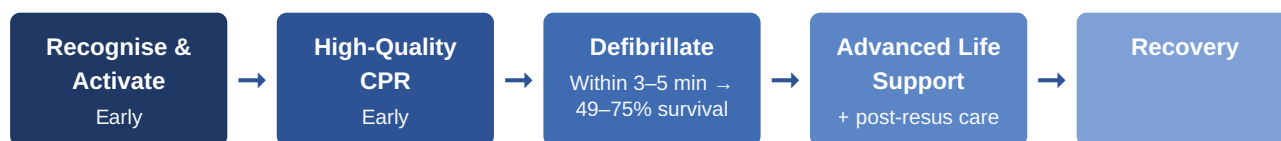
Complete chest
recoil

**Minimise**

Interruptions <10
sec

	Child — single rescuer	Infant — single rescuer
Check	Carotid pulse, scan chest 5–10s	Brachial pulse + resp, ≤10s
Technique	1 hand, lower sternum	2-finger, lower sternum
Ratio (trained)	15:2 ×5 cycles	15:2 ×10 cycles
Ratio (lay)	30:2 ×5 cycles	30:2 ×5 cycles
Depth	≈2 in	1.5 in / ½ AP

Multi-rescuer: R1 compresses, R2 airway + BVM breaths, R3 runs the AED — switch roles whenever the AED analyses.
 Infants: two-thumb encircling technique.



5. Crash Cart & Code Preparedness

Cart at foot of bed; one nurse manages it (drugs, flushes, syringes); document everything. Per-shift check: drawers sealed, drug expiry current, defib/BVM/O₂ functional. After a code: restock immediately, log use.

6. Post-Cardiac-Surgery Arrhythmias



Junctional Ectopic Tachycardia

Narrow-complex, AV dissociation, first 24–72 hr post-repair, poorly tolerated. Cool the patient, optimise sedation, involve cardiology early — avoid pro-automaticity agents.



Complete Heart Block

Can follow surgery near the conduction system. Keep temporary pacing wires & a pacing plan ready.

New post-op tachyarrhythmia: narrow vs wide QRS · check temp/pulse/BP/glucose/electrolytes · 12-lead ECG + repeat echo · inform cardiologist · prep drugs + defibrillator.



LINKS TO THE HANDS-ON LAB

Mega-code simulation, defibrillator/cardioversion practice, and post-cardiac-surgery rhythm-ID case stations — see the Workshop Schedule.

Compiled & adapted from

- PEERS Manual (2025) — Arrest Rhythms, Emergency Drugs, Crash Cart Structure (pp. 44–46); PALS appendix.
- BPCCN Guide — Identifying Rhythm Disturbances for Nurses, case-based ECG stations (pp. 46–52).
- PeC3N — post-operative cardiac arrhythmia case material.

Page-level citations to be added in a subsequent pass. Content has been edited and condensed for print/online study use; refer to the full source manuals for exhaustive detail.

05

Neurological Emergencies

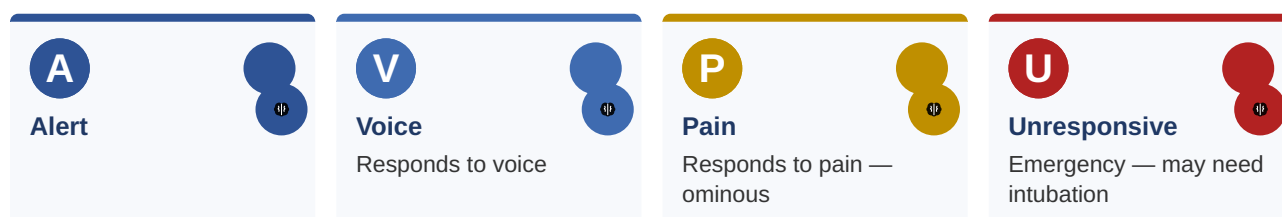
Manage seizures, recognize raised ICP, and implement cluster nursing.

5. Neurological Emergencies

Learning Objectives

- Score conscious level accurately using AVPU and the paediatric Glasgow Coma Scale.
- Run the status epilepticus algorithm within the correct time windows.
- Apply the 10 commandments of raised ICP management and deliver cluster care.
- Care safely for a child with an external ventricular drain (EVD), including recognising shift and drift.

1. Neurological Assessment



Check pupils (size/equality/reaction — unequal/dilated suggests raised ICP) and glucose in every drowsy child (<60 mg/dL = treat now).

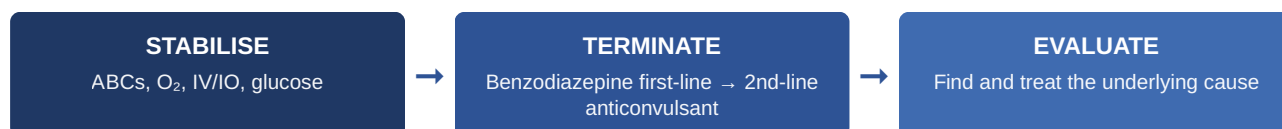
2. Status Epilepticus



DEFINITION

Seizure >5 min, OR repeated seizures without recovery in between. Any child brought in actively seizing = treat as status epilepticus.

SE type	Treat by (t1)	Long-term risk after (t2)
Tonic-clonic	5 min	>30 min
Focal, impaired awareness	10 min	>30 min
Absence SE	10–15 min	Unknown



3. Raised ICP — the 10 Commandments

1 Position Midline, 30–45° head-up	2 Cluster Care Group interventions, minimise stimulation	3 Sedo-analgesia Deep, adequate	4 Glucose <180 mg/dL	5 Sodium 145–150 mEq/L
6 PaO₂ 80–120 mmHg	7 PaCO₂ 35–40 mmHg	8 Hyperventilate 100% FiO ₂ only if herniation (unequal pupils)	9 Temp & Seizures Aggressive control	10 Hyperosmolar Rx 3% NS or Mannitol if herniation impending

CPP = Mean Arterial Pressure – ICP. Maintaining age-appropriate CPP matters as much as controlling ICP — avoid hypotension.

4. External Ventricular Drain (EVD) Care

 Infection Control Aseptic handling, monitoring, cleaning policy; assess for infection	 Alignment Maintain alignment with the tragus; know on/off stopcock positions	 Neuro Watch Use GCS to catch early deterioration
--	---	---

Term	Meaning
Shift	Significant catheter position change — often from movement/repositioning
Drift	Gradual, subtle displacement over time

5. Hypoglycaemia — the Common Contributor



TREAT & RECHECK

10% Dextrose 2.5 mL/kg IV (neonates) or 10 mL/kg (older children). Recheck glucose after treatment — recurrent hypoglycaemia needs investigation.



LINKS TO THE HANDS-ON LAB

GCS/AVPU scoring OSCE stations, seizure-management simulation, EVD care demonstration and cluster-care planning — see the Workshop Schedule.

Compiled & adapted from

- PEERS Manual (2025) — Management of Neurological Emergencies: Trauma, Seizures/Status, Hypoglycaemia (pp. 47–60).
- BPCCN Guide — Raised Intracranial Pressure, CPP Monitoring, EVD Competencies (pp. 77–80).

Page-level citations to be added in a subsequent pass. Content has been edited and condensed for print/online study use; refer to the full source manuals for exhaustive detail.

06

Trauma, Toxicology & Special Populations

Recognize hemorrhagic shock in the polytrauma patient and initiate MBTP; manage toxicologic emergencies; recognize and manage emergencies in the immunocompromised, oncology and solid-organ-transplant child.

6. Trauma, Toxicology & Special Populations

Learning Objectives

- Run a structured primary and secondary trauma survey with simultaneous assessment and stabilisation.
- Perform safe C-spine immobilisation, log-roll, and helmet removal.
- Recognise indications for gastric decontamination and activated charcoal, and manage organophosphorus poisoning.
- Calculate the Absolute Neutrophil Count and recognise febrile neutropenia and tumour lysis syndrome.
- Apply safe practice principles to blood transfusion and chemotherapy administration.

1. Approach to the Trauma Patient

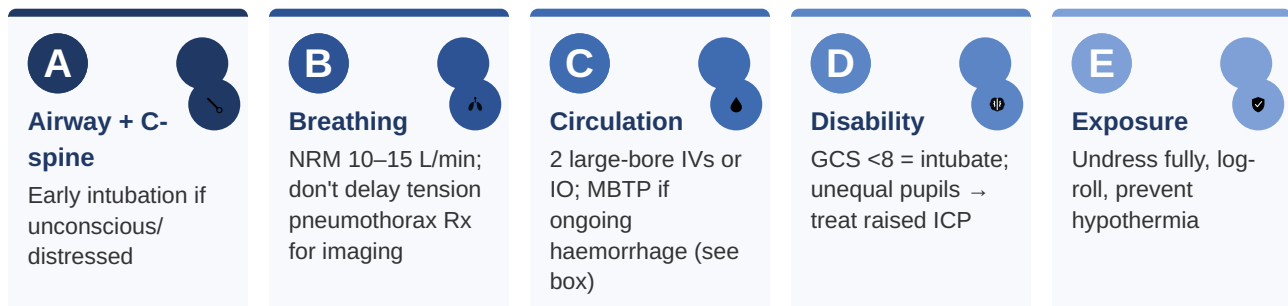


TEAM ROLES (5-6 PEOPLE, MINIMUM)

Team leader (foot end) · C-spine & airway (head end, 2 people) · Circulation/IV/samples/fluids
· Monitor/primary survey · AMPLE history/liaison with blood bank & radiology.

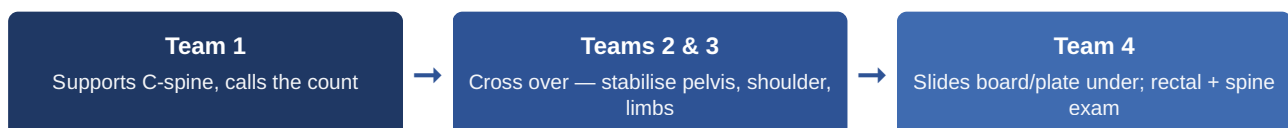
Trauma assessment and stabilisation happen **simultaneously**, not sequentially. Avoid head-tilt–chin-lift — use jaw-thrust with manual in-line C-spine control. Assume C-spine injury until proven otherwise.

Primary Survey — ABCDE with Haemorrhage Control



MASSIVE BLOOD TRANSFUSION PROTOCOL (MBTP)

Bolus crystalloid capped at 20 mL/kg, then blood products. O-negative if group unknown. Plasma:Platelets:PRBC = 1:1:1 if instability persists. Control external bleeding (pressure/tourniquet/pelvic binder); assume internal haemorrhage until excluded.



Secondary survey after primary + resuscitation: full AMPLE history, head-to-toe exam, selective imaging, E-FAST for internal bleeding.

2. Toxicology



Gastric Lavage

Only within 1 hr (up to 12 for some agents), airway secured. Contraindicated: unprotected airway, corrosive/hydrocarbon ingestion, >1 hr since ingestion.



Activated Charcoal

0.5–1 g/kg within 1 hr (4 hr for delayed-release/anticholinergic/opioid). Multi-dose: 1 g/kg q4h for long-half-life agents.

Muscarinic	Nicotinic	CNS
Diarrhoea, urination, miosis, bronchorrhoea, emesis, lacrimation, salivation, sweating	Weakness, fasciculations, paralysis, HTN, tachycardia	Confusion, seizures, ataxia, resp depression



OP POISONING — ATROPINISATION

Endpoints: clear lungs (most reliable) · HR>80 · SBP>80 · pupils not pinpoint · dry axillae · bowel sounds present. Dose: 0.05 mg/kg IV q5min until atropinised, then 20% of loading dose/hr infusion. + Pralidoxime: 20–40 mg/kg load then 500 mg/hr (no role after 48 hr). Poison Helpline (CMC Vellore): 1800-425-1213.

3. The Immunocompromised & Oncology Child



ABSOLUTE NEUTROPHIL COUNT

ANC = (Neutrophils % ÷ 100) × WBC. Example: WBC 4,000 × 60% = ANC 2,400. Neutropenia <1,500 · Severe <500 · Very severe <100.

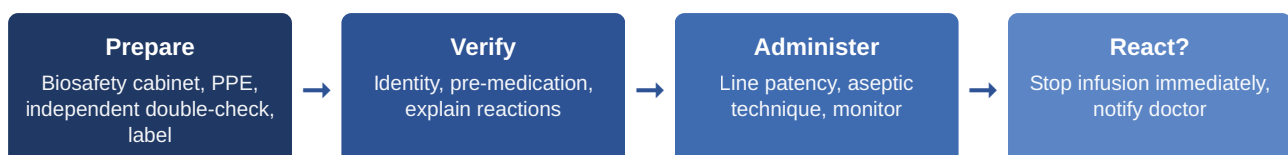
At-risk: 7–10 days post-chemo, post-BMT until recovery, genetic immune deficiency, new leukaemia/aplastic anaemia. Infection source: own flora (mouth, gut, perianal) + indwelling lines.



FEBRILE NEUTROPENIA

Single temp ≥38.5°C, OR ≥38°C sustained 1 hr, AND ANC <500 (or expected <500 in 48 hr). Only 50% have identifiable focus, cultures positive in 10–15% — treat empirically, broad-spectrum antibiotics ASAP. High-risk: ANC<100, abnormal CXR, expected neutropenia >10 days, toxic appearance, comorbidity, not in remission.

TLS warning signs	Lab abnormalities
N/V, cramps, fatigue, confusion, seizures, arrhythmia, dark urine	↑K, ↑PO ₄ , ↓Ca, ↑uric acid, ↑creatinine



4. HSCT & Solid Organ Transplant

Nurses watch closely through conditioning → infusion → engraftment → recovery for infectious/metabolic/mechanical emergencies during profound immunosuppression. For transplant recipients: monitor for rejection and infection, reinforce strict infection control.



LINKS TO THE HANDS-ON LAB (2-DAY)

Day 1: primary/secondary survey, C-spine/log-roll/helmet-removal, MBTP activation drill. Day 2: toxidrome case labs, burn-formula calculation, neutropenic sepsis/oncologic emergency stations — see the Workshop Schedule.

Compiled & adapted from

- PEERS Manual (2025) — Neuro/Trauma; Common Life-Threatening Ingestions; Toxidromes & OP Poisoning (pp. 47–68).
- BPCCN Guide — Approach to Polytrauma, Log Roll, Cervical Collar (pp. 72–76).
- CHAIN Workshop Manual (PediCritiCon 2025) — Care of the Neutropenic Child; Emergencies in Hemato-Oncology; Safe Practices in Blood Transfusion & Chemotherapy; HSCT & Solid Organ Transplant Nursing.

Page-level citations to be added in a subsequent pass. Content has been edited and condensed for print/online study use; refer to the full source manuals for exhaustive detail.

07

Safe Transport Practices

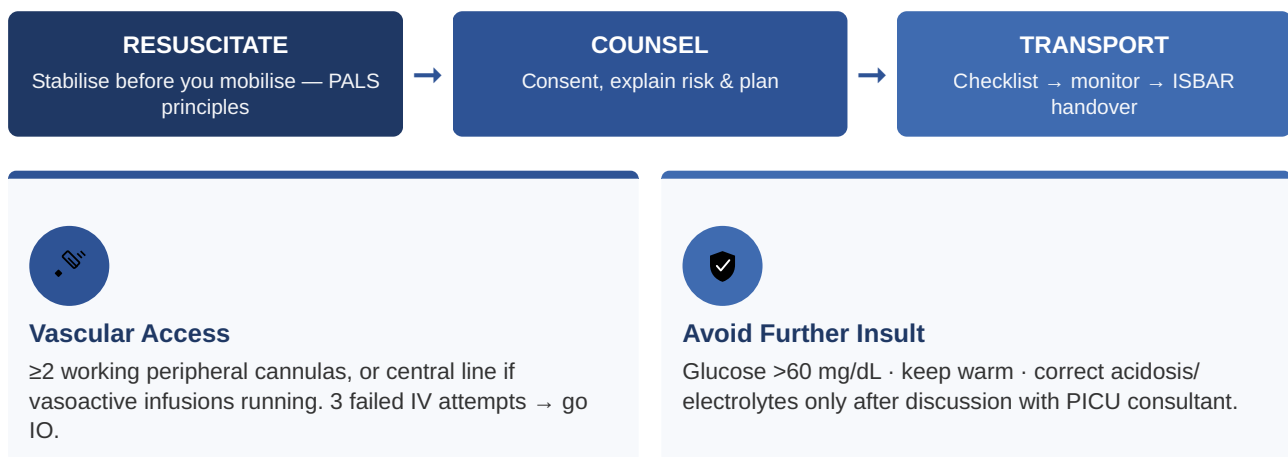
Stabilize before mobilizing, and execute safe intra- and inter-facility transport.

7. Safe Transport Practices

Learning Objectives

- Apply the resuscitate → counsel → transport sequence before any transfer.
- Pack and check a transport bag against a standard equipment checklist.
- Calculate oxygen cylinder requirements for a planned transport.
- Deliver a complete, structured ISBAR handover at the receiving unit.

1. The Transport Sequence



2. Counselling & Consent

Before transport, counsel on: illness nature/severity · risk of destabilisation/mortality in transit · likely plan on arrival · cost of care if asked. Document discussion, obtain signed consent before departure.

3. Oxygen Cylinder Calculation



ALWAYS CALCULATE FOR DOUBLE THE EXPECTED TIME

Spontaneous: $O_2 = \text{flow (L/min)} \times 60 \times \text{hours} \times 2$. Ventilated: $O_2 = \text{Minute Ventilation} \times 60 \times \text{hours} \times 2$.

Formula: Duration (min) = $\{(\text{Cylinder pressure} - \text{safe residual}) \times \text{factor}\} \div \text{flow rate (L/min)}$.

Cylinder	Factor
D-type	0.16
E-type	0.28
J-type	3.14

Worked example: E-type, 2000 PSI, residual 200 PSI, flow 5 L/min → $(2000-200) \times 0.28 \div 5 = 504 \div 5 \approx 100.8 \text{ min} (\approx 1 \text{ hr } 40 \text{ min})$.

4. Pre-Transport Checklist

A Airway/ Breathing Secure; O ₂ ×2 safety margin; working suction	C Circulation ≥2 IV/IO; pumps charged w/ battery; drugs drawn up + labelled	M Monitoring Portable monitor charged; alarms set	D Documentation Consent, records, handover note ready	T Team Roles assigned; receiving unit informed, bed confirmed
--	--	---	---	--

5. BLS vs ALS Ambulance

 BLS Ambulance Stable child, monitoring only	 ALS Ambulance Unstable, ventilated, or at risk of deterioration — airway/ vent/drug infusion capable
--	--

6. Handover — ISBAR

I Introduction	S Situation	B Background	A Assessment	R Recommendation
---------------------------------	------------------------------	-------------------------------	-------------------------------	-----------------------------------

WORKED ISBAR EXAMPLE

I: "Sister ... from ER." **S:** "Master Ayush needs an urgent PICU bed." **B:** "5-yr boy, septic shock, 3 days fever, 1 day lethargy." **A:** "Airway patent; SpO₂ 100% on NRM; HR 150, CRT 4s, BP 78/35 post-bolus, responds to voice, glucose 85. 2 IV lines, samples sent." **R:** "Shifting now — prepare bed, anticipate central line/intubation/catheter, give antibiotic when available."

LINKS TO THE HANDS-ON LAB

Transport-bag packing, a stabilize–mobilize simulated transfer, and an ISBAR handover OSCE — see the Workshop Schedule.

Compiled & adapted from

- BPCCN Guide — Transport of Sick Children: Resuscitation, Counselling & Consent, During Transport, O₂ Calculation, Pre-Transport Checklist, BLS/ALS Ambulance (pp. 63–69).

Page-level citations to be added in a subsequent pass. Content has been edited and condensed for print/online study use; refer to the full source manuals for exhaustive detail.

08

Renal, Metabolic & Extracorporeal Therapies

Monitor fluid balance, manage DKA, recognize hyperkalemia, and understand indications, setup and nursing care for renal replacement and extracorporeal life support (IHD, PD, CRRT, plasmapheresis, ECMO).

8. Renal, Metabolic & Extracorporeal Therapies

Learning Objectives

- Recognise and manage diabetic ketoacidosis and severe hyperkalaemia in a stepwise, safe manner.
- Compare IHD, PD, CRRT and SLED and identify indications for each.
- Describe the CKRT circuit, its core components, and common treatment modes (SCUF, CVVH, CVVHD, CVVHDF).
- Prepare for and monitor a therapeutic plasma exchange.
- Describe the two ECMO circulations (VA vs VV) and the core monitoring priorities for a child on ECMO.

1. Diabetic Ketoacidosis (DKA)



DIAGNOSTIC TRIAD

Glucose >200 mg/dL · venous pH <7.3 or HCO_3^- <15 mmol/L · ketonaemia/ketonuria.

1

Fluid resuscitation

Cautious isotonic bolus (10–20 mL/kg) only if shocked, then rehydrate evenly over 24–48 hr — rapid correction risks cerebral oedema



2

Insulin

Low-dose IV infusion 0.05–0.1 U/kg/hr, only AFTER 1st hour of fluids — never bolus in children



3

Potassium

Add once urine output confirmed & K^+ known — total body K^+ is depleted even if serum looks high-normal



4

Monitor

Hourly glucose + neuro status (cerebral oedema: headache, altered behaviour, bradycardia, HTN); 2–4 hourly electrolytes/gas

2. Hyperkalaemia — 3 Steps

STABILISE

IV calcium gluconate if ECG changes



SHIFT

Salbutamol neb/IV, insulin-dextrose, NaHCO_3 if acidotic



REMOVE

Diuretics if renal function OK; RRT if refractory/severe AKI

3. Choosing a Renal Replacement Modality

Modality	Best for	Key feature
IHD	Stable child, rapid clearance	3–4 hr sessions, 3×/week
PD	AKI/ESRD, limited infrastructure	Peritoneal membrane; CAPD/APD; low-cost
CRRT/CKRT	Unstable, critically ill	Slow, continuous, better tolerated
SLED	Intermediate acuity	Hybrid — longer sessions, lower flow



INDICATIONS FOR RRT

AKI with anuria/oliguria · overload unresponsive to diuretics · refractory hyperkalaemia · severe metabolic acidosis · uraemic symptoms · dialyzable poison · tumour lysis · crush syndrome.

4. CKRT — Mechanisms & Modes



Diffusion

Concentration gradient across membrane — removes small molecules (urea, creatinine)



Convection

Solvent drag under pressure — removes small–medium molecules + water



Ultrafiltration

Water removal by pressure gradient

Mode	Mechanism	Typical use
SCUF	Ultrafiltration only	Pure fluid overload
CVVH	Convection	Sepsis, overload, MOF
CVVHD	Diffusion	AKI/CKD, TLS
CVVHDF	Diffusion + convection	Liver failure, IEM

Circuit core: access/return pressure pods, filter pressure pod, blood pump, air-bubble detector, blood-leak detector, deaeration chamber, dialysate/replacement/effluent bags, anticoagulation syringe pump. Choice depends on access, clinician experience, hemodynamics, and fluid vs solute priority.

5. Therapeutic Plasma Exchange

	Centrifugal apheresis	Membrane separation
Pro	No heparin, peripheral IV OK	Faster, smaller, no citrate
Con	Large/expensive, needs citrate	Sieving-limited, needs HD catheter + heparin

**PRE/DURING-PROCEDURE MEDS**

Calcium (offsets citrate hypocalcaemia) · Hydrocortisone · Avil · Adrenaline available for anaphylaxis. Exchange volume $\approx 0.065 \times \text{wt(kg)} \times (1 - \text{Hct})$, 1–1.5 plasma volumes, 3–5 cycles.

6. ECMO — Bedside Essentials

**VV ECMO**

Respiratory support only, native heart pumps. SpO₂ goal >90% (>80% acceptable); pulsatile trace expected.

**VA ECMO**

Cardiac + respiratory support. Narrow pulse pressure expected; MAP target 45–70 mmHg; watch right-hand SpO₂ with femoral cannulation.

**IMMEDIATE LIFE THREATS**

Oxygenator/circuit clot · blocked cannula · raceway rupture · air embolism · accidental decannulation.

Monitor: neuro (consciousness, pupils, limb movement, NIRS) · resp (vent settings, ETCO₂, gas) · CV (rhythm, invasive BP, urine output, lactate, P1/P2/P3) · GI (feeds, bleed, distension).

**LINKS TO THE HANDS-ON LAB (2-DAY)**

Day 1: DKA calculation drills, CRRT/PD circuit familiarisation. Day 2: IHD/PD machine stations, plasmapheresis setup, ECMO circuit walkthrough/troubleshooting — see the Workshop Schedule.

Compiled & adapted from

- PEERS Manual (2025) — Common Metabolic Disorders in PICU.
- NEXT Workshop Handouts (PediCriticCon 2025) — Intermittent Hemodialysis; Peritoneal Dialysis; Continuous Kidney Replacement Therapy; Plasmapheresis; ECMO (pp. 1–56).

Page-level citations to be added in a subsequent pass. Content has been edited and condensed for print/online study use; refer to the full source manuals for exhaustive detail.

09

Infection Control & Quality

Bundle compliance to save lives, and drive continuous quality improvement using a structured point-of-care QI method.

9. Infection Control & Quality

Learning Objectives

- Apply the CLABSI, VAP and CAUTI prevention bundles correctly at the bedside.
- Perform hand hygiene per the WHO 5 Moments and 7-step technique.
- Apply antibiotic stewardship principles in daily practice.
- Run the 4-step POCQI method: identify a problem & write an aim statement, analyse & measure, test a change (PDSA), and sustain improvement.

1. Hand Hygiene & Standard Precautions



WHO 5 MOMENTS

Before touching a patient · before a clean/aseptic procedure · after body-fluid exposure risk · after touching a patient · after touching patient surroundings. Practice the 7-step handwash technique every time — it's a ritual, not an optional routine.

Precaution	When	Core measures
Standard	Every patient, every time	Hand hygiene, PPE per risk, safe injection, resp hygiene
Droplet	Flu, pertussis, mumps, meningitis, Hib, epiglottitis	Hand hygiene, surgical mask, private room/cohort

2. Device Disinfection — Spaulding Classification



Non-critical

Intact skin contact (stethoscope, BP cuff) → low/intermediate disinfection



Semi-critical

Mucous membranes (suction tubing, laryngoscope) → high-level disinfection



Critical

Sterile areas/vascular (surgical instruments, line kits) → sterilisation

3. Bundle Compliance

ABSI

Line bundle

Hand hygiene + max barrier at insertion · chlorhexidine · site selection · daily necessity review · scrub-the-hub

VAF

Ventilator bundle

HOB 30–45° · daily sedation interruption · oral care · subglottic suction · cuff pressure checks

AUTI

Catheter bundle

Insert only when indicated · aseptic technique · closed drainage · daily necessity review



ETT/LINE CARE REMINDER

Check ETT position every shift with a measuring tape, document in EMR. Suction only to the tip, never >0.5 cm beyond. Confirm wall-suction function; pressure –80 to –100 cmH₂O (max –120).

4. Antibiotic Stewardship

Narrowest effective spectrum, shortest effective duration, de-escalate promptly on culture results. Review every antibiotic order daily — a shared nurse–physician responsibility.

5. Quality Improvement — POCQI 4-Step Method

1

Identify + Aim Statement

Find real problems from local data, build a small multidisciplinary team, write what/whom/how much/by when



2

Analyse & Measure

Map the process, find root causes (not symptoms), choose measures



3

Test Changes (PDSA)

Plan–Do–Study–Act small cycles before scaling up



4

Sustain

Standardise into routine practice, keep monitoring, spread the change



WRITING A GOOD AIM STATEMENT

"Reduce time to first antibiotic dose for febrile neutropenia from 90 to 60 minutes for all admissions, within 3 months." — state what, whom, how much, by when.

6. Incident Reporting

Report every near-miss and adverse event without blame — it's the raw material for the next PDSA cycle. Ask "why" repeatedly until you reach a fixable system cause, not just an individual error.

**LINKS TO THE HANDS-ON LAB**

Live CLABSI/VAP/CAUTI bundle audits, hand-hygiene competency check, and a full POCQI case-based exercise (aim → root-cause → PDSA) — see the Workshop Schedule.

Compiled & adapted from

- BPCCN Guide — Infection Control, Disinfection & Sterilisation, Biomedical Waste Segregation, Standard/Droplet Precautions (pp. 34–41).
- EQUIP Learners' Manual (WHO-SEARO POCQI, 3rd Edition, 2020) — the 4-step Quality Improvement method.

Page-level citations to be added in a subsequent pass. Content has been edited and condensed for print/online study use; refer to the full source manuals for exhaustive detail.

PCCN COURSE · ONLINE MATERIAL

10

Medication Safety

Zero-harm medication practice.

10. Medication Safety

Learning Objectives

- Classify a medication error as prescription, transcription, administrative, dispensing, or monitoring.
- Apply the '5 Rights' and independent double-check to every high-alert medication.
- Recognise look-alike/sound-alike (LASA) drug pairs and build safeguards against confusing them.
- Report medication errors and near-misses without blame, as part of a learning system.

1. What Counts as an Error

A **near miss** is an unplanned incident with potential to cause harm that didn't — e.g. a pharmacist catching a dose error before dispensing. Near misses matter as much as real errors: they're free lessons.

2. Five Types of Medication Error — Real Cases



Prescription

Combo antibiotic ordered without specifying which component the dose refers to (e.g. Amox-Clav, Pip-Taz). Always confirm the component.



Transcription

4-yr-old, K⁺ 3.3: KCl 10 mEq ordered, nurse gives 10 mL instead — tachyarrhythmia. mEq vs mL confused.



Administrative

1-yr-old needing nebulisation given the drug IV instead — tachyarrhythmia. Route confusion.



Dispensing

Pre-extubation steroid planned; IV Dexmedetomidine given instead of Dexamethasone — bradycardia/hypotension. LASA name mix-up.



Monitoring

Peripheral dextrose line also used for phenytoin/Ca/Mg — swelling, ulcer. Incompatible infusions, one line, no site monitoring.

3. Steps to Avoid Errors

W

Write clearly
CAPITALS, no informal abbreviations; countersign every change

V

Verify identity
Confirm patient before entering data; default dose/route/frequency safeguards

L

LASA safeguards
Verify accuracy, label clearly, avoid frequent brand changes

D

Double-check
Independent double-check for every high-alert medication

R

5 Rights
Patient · Drug · Dose · Route · Time

S

Sterile cockpit
Reduce distraction/interruption during drug rounds

Standardise preparation/labelling · use tech (e-prescribing, barcode scanning, smart pumps) · build a blame-free reporting culture that surfaces problems early.

**HIGH-ALERT MEDICATION REMINDER**

Insulin, opioids, anticoagulants, concentrated electrolytes (KCl, hypertonic saline), sedatives/paralytics, chemotherapy — always independent double-check, no exceptions under time pressure.

**TAKE-HOME**

Every one of the five error types above is preventable with a consistent, unhurried, double-checked process.

**LINKS TO THE HANDS-ON LAB**

High-alert medication double-check drills, the "Stop the Drop" live simulation, and a LASA medication identification station — see the Workshop Schedule.

Compiled & adapted from

- BPCCN Guide — Medication Errors, Types & Case Examples, Steps to Avoid Errors (pp. 42–45).

Page-level citations to be added in a subsequent pass. Content has been edited and condensed for print/online study use; refer to the full source manuals for exhaustive detail.

11

Professionalism, Communication & Ethics

Crisis Resource Management (CRM) and communication principles across the care continuum.

11. Professionalism, Communication & Ethics

Learning Objectives

- Use ISBAR for every handover and AIDET for every family/patient introduction.
- Apply the SPIKES framework to breaking bad news.
- Recognise the red flags of child maltreatment and follow mandatory-reporting duty.
- Apply core CRM principles — closed-loop communication, shared mental model, situational awareness — in a code or crisis.

1. AIDET — Introducing Yourself

A

Acknowledge

Eye contact,
correct name,
show care

I

Introduction

Your name, role,
invite involvement

D

Duration

Expected time of
any procedure

E

Explanation

What & why, in
plain language

T

Thank You

For time and
cooperation



SMILE — TALKING WITH A CHILD

Speak softly · Meet at eye level · Involve the child · Let them express · Explain using toys and pictures.

2. ISBAR — Structured Handover

I

Introduction

Who you are, your
role

S

Situation

Why calling /
what's needed
now

B

Background

Relevant history

A

Assessment

Your ABCDE
findings +
interventions

R

Recommendation

What you need, by
when

Use for every shift handover, escalation call, and transfer — consistency prevents information loss (worked example in Section 7).

3. SPIKES — Breaking Bad News

1

Setting

Private space, sit down, minimise interruptions



2

Perception

Ask what the family already understands



3

Invitation

Ask how much detail they want now



4

Knowledge

Clear, jargon-free, small chunks



5

Emotion

Acknowledge and respond with empathy



6

Summarise

Recap, check understanding, agree next step

4. Crisis Resource Management (CRM)



Closed-Loop Communication

Order given → repeated
back → confirmed



Shared Mental Model

Everyone knows the
plan, role, and "success"



Situational Awareness

Continuously scan
patient, team,
environment



Clear Roles & Call-outs

Named leader; findings/
actions said out loud

5. Child Protection



SUSPECT NON-ACCIDENTAL INJURY WHEN...

Inconsistent/changing history · mechanism doesn't match injury · injuries at different healing stages · injury in a non-mobile infant · delayed presentation. Fully expose every sick/injured child for bruising/petechiae in unusual sites. Document objectively, follow mandatory reporting pathway.

6. End-of-Life & Ethics

Core principles: **autonomy** (respect wishes/consent), **beneficence/non-maleficence** (best interest, avoid harm), **justice** (fair resource allocation). Handle consent, DNR, and organ donation with SPIKES-based compassion, clearly documented.

7. FAST HUGS IN BED PleaSe — Daily Systematic Review

F Fluid therapy Type/rate, I&O balance	A Analgesia & Antibiotics Drug/dose/volume	S Sedation/ Secretions/ SBT Spontaneous breathing trial	T Thromboprophylaxis	H Head-end elevation 30°
U Ulcer prophylaxis	G Glucose control	S Skin Intact / bed sores / score	I Indwelling lines/ infection	N NG tube In situ / feeding regimen
B Bowel care	E Eyes & Environment Care / temp control / isolation	D De-escalation Wean vent/ inotropes/DNR review	P Physiotherapy Mobilisation	S Social support Insurance/billing



LINKS TO THE HANDS-ON LAB

AIDET/ISBAR role plays, a SPIKES simulated family scenario, a child-protection documentation case, and closed-loop CRM drills — see the Workshop Schedule.

Compiled & adapted from

- BPCCN Guide — AIDET: How to Introduce Yourself; Documentation, Teamwork & Communication; ISBAR Handover (pp. 5–9, 23–25).
- PEERS Manual (2025) — DIRECT Approach & handover principles.

Page-level citations to be added in a subsequent pass. Content has been edited and condensed for print/online study use; refer to the full source manuals for exhaustive detail.

12

Advanced Procedures, Mechanical Ventilation & PICU Liberation

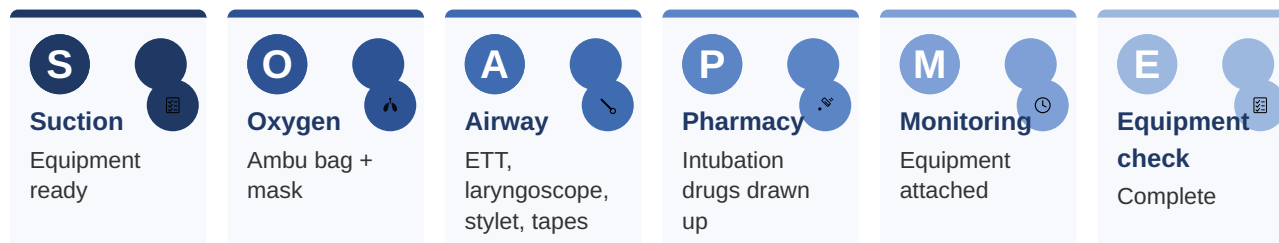
Safe preparation for intubation, central lines and arterial monitoring; deliver evidence-based invasive ventilation across disease states; lead sedation, delirium screening and early mobilisation to safely liberate the child from the ventilator.

12. Advanced Procedures, Mechanical Ventilation & PICU Liberation

Learning Objectives

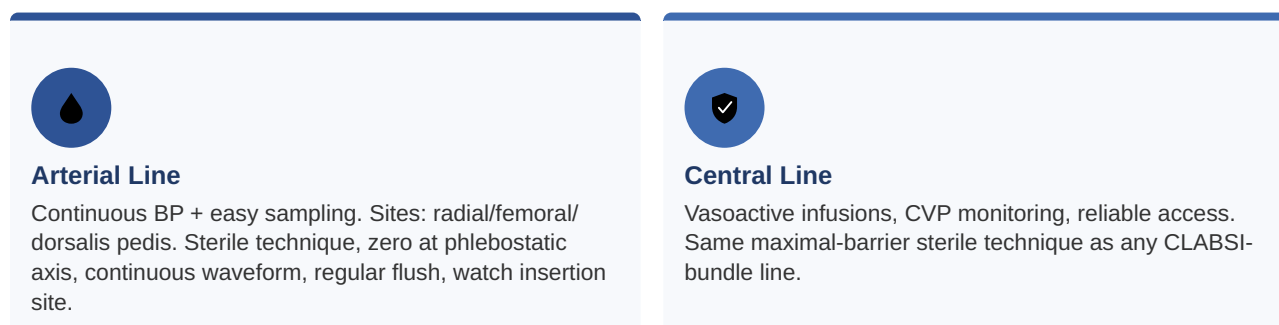
- Prepare a SOAPME-ready difficult-airway trolley and assist a safe intubation.
- Set up and troubleshoot an arterial line and interpret basic waveforms.
- Explain trigger/limit/cycle, common ventilator modes, and the key parameters to monitor.
- Run the DOPE algorithm for a suddenly desaturating ventilated child.
- Screen for pain and delirium and apply the PICU Liberation bundle to enable safe early mobilisation.

1. Preparing for Intubation — SOAPME

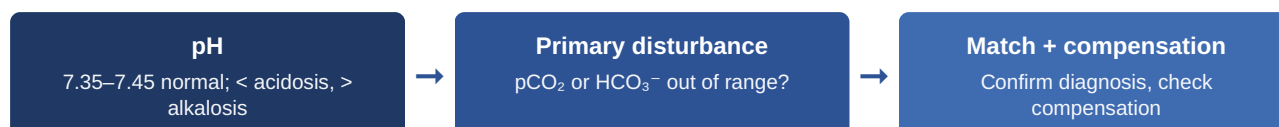


Keep an intubation tray ready at every ventilated child's bedside, always. Use Mallampati class to anticipate a difficult airway. Position for age: infants — shoulder roll, "sniffing position"; older children — head extension.

2. Vascular Monitoring — Arterial & Central Lines



ABG Interpretation — 3 Steps



Preferred puncture: radial artery (best collaterals) — Modified Allen's test first. Pre-heparinised syringe, 30–45°, firm pressure ≥5 min after.

3. Ventilation Fundamentals



EVERY BREATH HAS 3 PARTS

Trigger (time/pressure/flow — starts it) · **Limit** (volume/pressure — caps it) · **Cycle** (time/flow — ends it). The combination defines the mode.

Mode	Description
CMV/AC	Full support, every breath preset
SIMV	Synchronised + spontaneous breaths between; weaning
PSV	Supports patient-initiated breaths only — weaning
CPAP	Continuous pressure, spontaneous breathing
BiPAP	IPAP ventilates, EPAP prevents collapse
PRVC	Guarantees tidal volume at lowest pressure



Tidal Volume

6–8 mL/kg



PEEP

3–5 cmH₂O



I:E Ratio

Commonly 1:2



Suction Cath

ETT ID ×2

Monitor: SpO₂ >94% · EtCO₂ 35–45 · ABG · chest rise symmetry · bilateral breath sounds · HR/BP · alarms.

Alarm	Cause	Action
High pressure	Obstruction, kink, cough	Check tube, suction, adjust
Low pressure	Leak, disconnection	Reconnect, check cuff/circuit
Low volume	Disconnection, weak breaths	Check asynchrony, optimise sedation



THE DESATURATING VENTILATED CHILD — DOPE

Displacement · Obstruction · Pneumothorax · Equipment failure. Disconnect, hand-bag with 100% O₂ while working through DOPE. Airway trolley ready for re-intubation.

4. Positioning, Suctioning & Tracheostomy



Positioning

Neutral/slightly extended head;
HOB 30–45°; reposition q4h



Watch for

Sudden desat, ↑WOB, absent chest
rise, abnormal EtCO₂



Trach care

Spare + size-down tube at bedside
always; document size; ALWAYS 2
people present

5. PICU Liberation Bundle

Pain/sedation/delirium assessment + minimisation, paired with early mobilisation, shortens ventilation and reduces delirium — safely, with a structured checklist.



Wong-Baker FACES

Self-reported, ≥3 years



FLACC

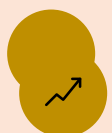
Observer-rated, 2mo–7yr or non-verbal; 0–10, 5 criteria

Delirium screening: psCAM-ICU (<5yr dev age), pCAM-ICU (>5yr), CAPD (0–18yr), SOS-PD.



NON-PHARMACOLOGIC DELIRIUM MANAGEMENT

Sleep hygiene · interdisciplinary rounds · family engagement · early mobilisation. Minimise benzodiazepine-based sedation wherever feasible.



EVIDENCE FOR EARLY MOBILISATION

Adult ICU meta-analysis: 47% lower delirium risk, 1.8 days shorter duration. LiberAction PICU bundle: mobilisation adherence ≈70%, delirium down to 23%, shorter ventilation/PICU stay — no mobilisation-related adverse events.

Pre-Mobilisation Daily Huddle

1

Ready for mobilisation?

Patient identified



2

No contraindication

Active seizures, major bleeding, unstable fracture, surgical emergency



3

Senior clearance

Assessed by resident/consultant



4

Team alignment

Goals reviewed with physio, nursing, family



5

Resources ready

Personnel + equipment available



6

Lines/airway secure

Dressings intact



7

Analgesia optimised

Proceed only if ALL answers are Yes

**LINKS TO THE HANDS-ON LAB (2-DAY)**

Day 1: SOAPME/RSI prep, difficult-airway/LMA/cricothyrotomy/needle-decompression stations, central/arterial line insertion. Day 2: NIV/invasive vent setup, waveform troubleshooting, DOPE drill, trach care, sedation/delirium screening, PICU Liberation workstations — see the Workshop Schedule.

Compiled & adapted from

- PEERS Manual (2025) — Preparation for Procedures in PICU, O2 Delivery Devices, Intubation, LMA (pp. 15–21).
- BPCCN Guide — Preparation for Intubation, Mallampati Score, Arterial Line, ABG Interpretation (pp. 32–33, 53–60, 70–71).
- BMV Workshop Manual (PediCritiCon 2025) — Invasive Ventilation, Modes & Waveforms, Troubleshooting, Tracheostomy Care (pp. 19–70).
- LEAP Handout (PediCritiCon 2025) — PICU Liberation Bundle, Pain & Delirium Assessment, Early Mobilisation (pp. 6–21).

Page-level citations to be added in a subsequent pass. Content has been edited and condensed for print/online study use; refer to the full source manuals for exhaustive detail.